



FIGURE 14.2 EUR exchange rate and 5Y basis swap spread between 3M EURIBOR and 6M EURIBOR.
Sources: data – Bloomberg; chart – Authors.

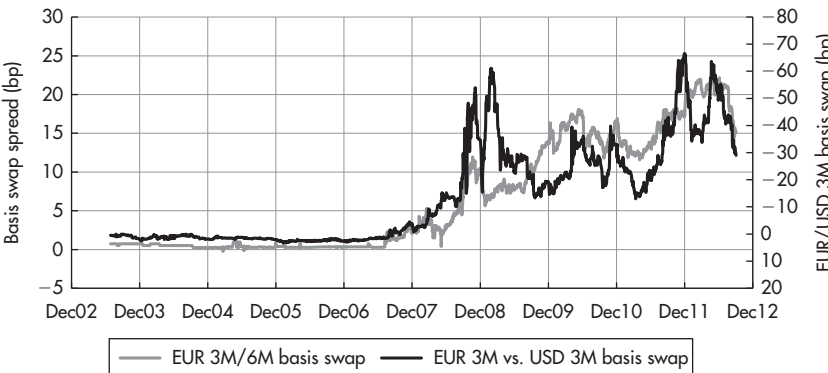


FIGURE 14.3 5Y basis swap spread between 3M EURIBOR and 6M EURIBOR and 5Y basis swap between 3M EURIBOR and 3M USD LIBOR.
Sources: data – Bloomberg; chart – Authors.

the correlation between these two variables has been -0.62. (The convention is to quote the EUR 3M vs. USD 3M basis swap as a spread to the EUR leg, and the cross-currency basis swap is shown on an inverted axis in Figure 14.3.)

The correlation between ICBS, especially those covering the secured–unsecured basis, and cross-currency basis swaps (CCBS) has been stable over the years, reflecting the link via a causal relationship: in times of higher (perceived) credit risk, for example, due to stress in the banking sector, the

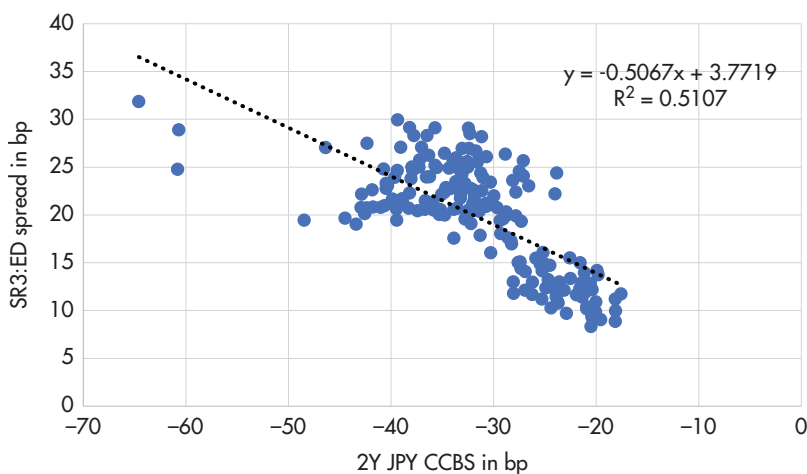


FIGURE 14.4 Regression of the 3M SOFR versus Eurodollar future spread against the 2Y USD–JPY cross-currency basis swap.

Source: Huggins and Schaller (2022).

preference for both secured versus unsecured lending and for keeping funds in the US banking system increases. Exploiting this relationship shown in Figure 14.4, Chapter 4 of Huggins and Schaller (2022) presents a model for the ICBS (as traded in SOFR versus Eurodollar or Fed Fund futures) relative to the CCBS. One application of this model is the possibility of replacing the capital-intensive CCBS in some relative value (RV) trades (which do not require the actual exchange of principal) with cheaper spread future contracts.

Chapter 15 will use the secured–unsecured basis as one input variable for pricing CCBS – as this chapter does for the ICBS and Chapter 12 did for swap spreads.

ROLE AS BUILDING BLOCKS

ICBS are of interest in their own right, as instruments for hedging various transactions, as a means for trading perceptions of credit risk in the banking sector and as a market price for the secured–unsecured basis. In addition, they fulfill an essential role as a building block for other structures, primarily bonds that are cross-currency basis swapped into other currencies.

For example, if a German Bund asset swapped into 6M EURIBOR should be basis swapped in USD, in a first step, an ICBS exchanges the 6M EURIBOR into ESTR payments, which are then exchanged in a second step by a CCBS into USD SOFR. There are also quotes for CCBS offering a direct exchange between EURIBOR and SOFR, but their pricing (and usually also their execution) does involve the term (risk) given by the EURIBOR–ESTR ICBS.

After CCBS have been treated in Chapter 15, ICBS will then be used in this manner as building blocks for the construction of combinations between asset, basis and credit default swaps occupying Chapter 16.

Cross-Currency Basis Swaps

DEFINITION

A cross-currency basis swap (CCBS) exchanges two reference rates in different currencies. In terms of Table 11.1, it therefore links two different rows. In practice, all CCBS have USD as one leg; for example, a CCBS between ESTR and TONAR is executed as a combination of two CCBS, one between ESTR and SOFR and one between SOFR and TONAR. The transition from LIBOR to SOFR in the US therefore resulted in CCBS now (almost) always¹ involving different types of reference rates as well, the secured O/N rate SOFR in USD and an unsecured (O/N or term) rate in EUR, JPY, or GBP. In terms of Table 11.1, this corresponds to the CCBS involving a switch between both rows and columns.

This chapter starts with outlining the situation before the transition, i.e. assumes LIBOR as reference rate in all currencies, and then explains the impact of the transition on the instruments and their pricing.

FX Swap

In a spot FX transaction, two parties exchange two currencies immediately upon agreement of the exchange rate.

In a forward FX transaction, two parties agree on an exchange rate and on a date in the future on which the two currencies will be exchanged.

The relation between the spot exchange rate and the forward exchange rate typically precludes arbitrage opportunities, in accordance with the standard equation:

$$F_{\frac{USD}{EUR}} = \frac{1 + R_{USD}}{1 + R_{EUR}} S_{\frac{USD}{EUR}}$$

¹Being an OTC product, CCBS between SOFR and the ECB policy rate, i.e. another secured reference rate, are theoretically conceivable, but do not appear to play a major role in the market.

where

$S_{\frac{USD}{EUR}}$ is the number of US dollars that can be exchanged for one euro in the spot market;

$F_{\frac{USD}{EUR}}$ is the number of US dollars that can be exchanged for one euro three months forward;

R_{USD} is the *non-annualized* rate at which a bank could lend US dollars on an unsecured basis for three months;

R_{EUR} is the *non-annualized* rate at which a bank could borrow euros on an unsecured basis for three months.

If this condition were not satisfied, one could make arbitrage profits by borrowing euros, buying dollars in the spot market, lending the dollars, and selling the dollars forward – or by doing the opposite, depending on whether the forward rate was higher or lower than the forward exchange rate implied by this no-arbitrage equation.

The FX swap is a combination of the spot and forward transactions discussed above. In other words, the FX swap consists of the spot and forward transactions by which one would attempt to exploit the arbitrage in the event the equation above was not satisfied.

It's important to note the difference between an FX swap and an FX forward. In an FX forward, the two currencies are exchanged only on the forward date. In an FX swap, currencies are exchanged on both the spot date and on the forward date. As we'll see later in the chapter, the distinction is important for understanding the economic function of the FX swap, as it eliminates the FX exposure.

An FX swap can be conceptualized as consisting of three components:

1. an initial exchange of principal in two currencies;
2. a return at the swap end date of the principal exchanged at the start date;
3. a payment of interest in each currency at the swap end date, added to the principal returned.

Because the interest payments in step (3) typically differ in value, the exchanges of principal and interest payments at the end date of the swap constitute an effective forward rate that typically is different from the spot exchange rate.

FX swaps typically have tenors of up to one year. When the principal is to be exchanged for a period greater than a year, a CCBS is more typical. In fact, the CCBS can be thought of as an extension of the forward FX market to longer maturities – just as we have thought of the asset swap spread as a repo–LIBOR basis swap, i.e. an extension of the repo market to longer maturities.

Cross-Currency Basis Swaps

A CCBS is very similar to an FX swap in that it involves:

1. an initial exchange of principal in two currencies;
2. the return of this principal at the end date of the swap;
3. the interest payments in each currency to reflect the time value of the funds borrowed by each counterparty in the swap.

Where an FX swap involves payment of a single interest payment in each of the two currencies, a CCBS involves a stream of interest payments in each of the two currencies. Usually, both legs have floating payments. For example, before the transition a five-year (5Y) EUR/USD CCBS might involve quarterly payments of three-month (3M) USD LIBOR in exchange for quarterly payments of 3M EURIBOR. After the transition, it might involve quarterly payments of the compounded USD SOFR rate in exchange for quarterly payments of the compounded EUR ESTR rate. Being an OTC product, also fixed payments on one or both legs of the CCBS are conceivable. However, it is much more common to trade a CCBS with two floating legs and then add one (or two) (asset) swap between floating and fixed, i.e. to keep all (asset) swaps between floating and fixed free of currency exchanges. This book follows the market convention (and liquidity) and in Chapter 16 treats swaps between fixed rates in one currency and floating rates in another currency as a combination of a CCBS between two floating rates and an asset swap without currency exchange.

The key to understanding the CCBS is to keep in mind the three components of the swap:

1. an initial exchange of principal in each currency;
2. the return of principal in each currency at the end of the swap;
3. the interest rate payments during the swap paid to each counterparty to reflect the time value of money in that currency.

In these examples, we've discussed fixed and floating payments without reference to any spreads, while in practice CCBS typically are traded with an attached spread to reflect the balance of supply and demand. For example, one swap counterparty might agree to pay USD LIBOR in exchange for EURIBOR less a spread in order to provide sufficient motivation for a counterparty to engage in the swap. Usually, the "CCBS spread" is referred to simply by the name for the instrument and is called "CCBS" as well.

As a practical matter, the spread could be attached to either leg of the swap. For example, the same counterparty might agree to pay USD LIBOR

plus a spread in exchange for EURIBOR. By convention, CCBS spreads tend to be quoted with reference to the non-dollar leg when the swap involves USD. For example, a 5Y CCBS quote of $-38/-37$ would mean that an investor could exchange dollars for euros by agreeing to pay 3M EURIBOR less 37 basis points (bp) and to receive 3M USD LIBOR for five years – or the investor could choose to receive 3M EURIBOR less 38 bp in exchange for 3M USD LIBOR for five years.

APPLICATIONS OF THE CCBS

As with any new asset class, it can be analyzed in isolation from others and screened for opportunities by statistical assessment of its intrinsic relationships. We have illustrated this in the case of CDS in Chapter 13 and in principle the same methods can be applied to the CCBS market as well. For example, the term structure of the CCBS curve(s) can be subjected to a PCA. While statistical analysis is always an important element of a comprehensive insight, the function of the CCBS to link the funding rates of different markets results in it being highly dependent on the flows between these markets and hence a bad candidate to be traded via statistical tools only. We will therefore provide an overview over some applications of the CCBS before discussing their impact on its pricing.

CONSTRUCTING ANY REFERENCE RATE FROM ANY OTHER

Via a combination of CCBS and ICBS, any reference rate globally can be swapped into any other. Once more in terms of Table 11.1, any cell can be connected with any other cell.

If one considers USD SOFR as the global benchmark due to it combining the safe haven currency with the safest reference rate practically possible, one can construct from this starting point all other USD reference rates via ICBS, all EUR reference rates via a USD–EUR CCBS (maybe together with a EUR ICBS), all JPY reference rates via a USD–JPY CCBS (maybe together with a JPY ICBS), etc. For example, in order to transform SOFR into EURIBOR, one could either use directly a USD SOFR–EURIBOR CCBS or alternatively a USD SOFR–EUR ESTR CCBS together with a ESTR–EURIBOR ICBS within EUR. Actually, even if one chooses the first option, it is likely that the broker will execute the trade via the second option.

Or taking the perspective of a German corporate funding at EURIBOR plus a spread, it can use the market quotes for ICBS and CCBS to calculate its funding costs against any reference rate worldwide. And if it finds that it can fund cheaper against another rate than against EURIBOR, the basis swaps enable exploiting this difference.

ISSUING FOREIGN BONDS WITHOUT FX EXPOSURE

An important function of a CCBS is to allow someone to engage in a transaction in a currency other than his domestic currency without taking FX risk. For example, an Italian issuer might issue bonds in USD and swap them into EUR via a CCBS in order to take advantage of a relatively large demand for his bonds among US investors. In so doing, he may be able to achieve a lower borrowing cost than he could achieve by issuing bonds in EUR.

In this example, there's nothing to prevent the issuer from simply selling bonds denominated in USD and then converting the proceeds to EUR, without engaging in any additional swaps. But in that case, the issuer is exposed to FX risk with respect to making payments of principal and interest in USD. To eliminate this FX risk, the issuer can engage in a CCBS immediately after issuing the bonds. The issuer receives EUR at the start of the swap, pays EUR at the end of the swap (equal to the maturity date of the bonds), and makes ongoing payments of EUR between the start and end dates.

Through the CCBS, issuers of debt can choose the currency in which they fund, independent of the denomination of their debt. For example, if an A-rated company funds in USD at LIBOR + 250 bp and in JPY at LIBOR + 150 bp, it has an incentive to issue Samurai bonds and to exchange its JPY into USD funding via the CCBS. If the CCBS were zero, it would realize about 100 bp funding advantage. Of course, the activity of Samurai issuers seeking to profit from this opportunity would be expected to pressure CCBS spreads and/or to widen JPY credit spreads so as to reduce the attractiveness of the opportunity.

INVESTING IN FOREIGN BONDS WITHOUT FX EXPOSURE

On the other side of the supply/demand balance, buyers of bonds can also choose the currency in which they invest, through the CCBS. For example, if a US Treasury bond trades at USD LIBOR - 50 bp and a Bund at EUR LIBOR - 20 bp (and both issuers are considered of the same credit risk), a USD-based investor might want to use the CCBS to obtain access to the EUR bond market without FX exposure. If the CCBS was equal to zero, he would realize a return of roughly USD LIBOR - 20 bp, thus about 30 bp pick-up versus the US Treasury investment. Now, the activity of investors seeking to take advantage of that opportunity will drive the CCBS up and/or Bund swap spreads wider (more negative) relative to US Treasury swap spreads.

The CCBS level observed in the market therefore reflects the sum of global capital flows chasing the currency which offers superior funding and investment opportunities. For example, high credit spreads in USD relative to JPY

could lead to an increase in Samurai issuance, driving the CCBS negative. This in turn might make basis swapped JGBs attractive on a USD LIBOR spread basis, generating flows that push the CCBS up again. Consequently, the pricing of the CCBS reflects the complex mixture of global funding and investment flows and can serve as a useful and quantitative indicator for these activities, which are often hidden from direct observation. Stated otherwise, the CCBS condenses the multitude of all cross-currency funding and investment flows into one easily observable and tradable number.

In abstract terms, basis swaps link global funding rates and therefore allow investors and borrowers to exploit global credit spread differences. And in a similar manner, global credit spread differences impact the level of the CCBS via global credit spread arbitrage.

Further impacts on the CCBS can arise from:

- Differences in yield curve steepness (carry) in different currencies. For example, a high carry difference between the Treasury and JGB yield curve typically prompts Japanese investors to increase their foreign bond holdings and to roll the FX hedge on a short-term (e.g. 1Y) basis. This mismatch between the maturity of the bond and of the CCBS results in a synthetic carry position along the US yield curve. Thus, whenever these carry positions are particularly attractive, demand for the short end of the CCBS curve tends to be affected.
- Hedging flows from structured products. Many structured products involve several currencies, and hence dealers require CCBS positions to hedge them. For example, the hedging flows of power reverse dual callables (PRDCs) introduce a link between the CCBS and the FX markets. When the FX rate moves, in particular through certain levels used in PRDCs, PRDC hedges need to be adjusted. Since most PRDCs are ultralong issues, this tends to affect 20Y and 30Y CCBS in particular.

Selecting the Banking System to Generate Money Market Returns

Imagine a USD-based institution that wants to invest its USD funds on a rolling 3M basis at the prevailing rate in the money market over the next five years. The most obvious possibility to achieve this modest goal is to simply choose every quarter a US LIBOR panel bank as counterparty to lend to. But the CCBS offers another way of achieving the same goal: enter into a 5Y CCBS, in this example the USD–EUR one, which exchanges the USD funds at the beginning into EUR. Then choose every quarter a EURIBOR panel bank as counterpart to lend the EUR funds to. This will generate EURIBOR cashflows, which the CCBS exchanges back into USD LIBOR cashflows. And at the end of the five years, the CCBS will also exchange the EUR principal back into USD at a

pre-defined exchange rate. Hence, this second option seems like a complicated way to achieve the same result: every quarter a USD LIBOR income and after five years the same USD funds originally invested (also in case of using the CCBS without FX risk). But there are a few differences:

- Most importantly, while the final cashflows are the same, they are generated via different counterparties: US LIBOR panel banks in the first and EURIBOR panel banks in the second option. If the first are considered to be safer, there is higher demand for the first option.
- The CCBS spread subtracted from the EURIBOR payments. Together with the statement of the first point, the CCBS spread can therefore be considered as market price for the “privilege” of keeping funds in the US banking system rather than lending to counterparties abroad. If there was no preference, the CCBS spread should be zero from this perspective. The more demand for generating money market returns in the US relative to other countries, the more negative the CCBS spread, as compensation for those choosing the second option. Correspondingly, it shows a clear correlation to stress in the banking sector: the CCBS tends to become more negative whenever bank risk is deemed to increase. Interestingly, this was also true when the problems of the banking sector originated in the US during the subprime crisis: after a few days of slightly positive USD–EUR CCBS spreads, significantly negative values followed soon and persisted for long. This is in line with the perception of the US banking system as a relative safe haven for any sort of crisis.
- The second option also involves higher transaction costs, capital requirements and counterparty risk – not only from exposure to foreign banks to generate the money market returns but also from exposure to the counterpart of the CCBS, from whom the investor expects to get the USD principal back after five years. As the principal is exchanged in a CCBS, this results in a significant exposure to the counterpart: imagine the counterpart in the CCBS is a European bank and defaults during a euro crisis. The investor is then left with a depreciated EUR principal; while the CCBS has pre-defined the FX rate, the counterparty no longer exists. Collateral and margin agreements are therefore of particular importance for (the pricing of) CCBS.

PRICING OF THE CCBS

Summarizing the driving forces of the CCBS (other than intrinsic statistical relationships), it reflects at the same time:

- The sum of global capital flows chasing the currency which offers superior funding and investment opportunities; *and*